

ENGINEERING

Nexora Platform API Documentation

REST API v2 reference: authentication, endpoints, rate limits, pagination, webhooks and errors

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Applies to	Internal engineers and integration partners	Classification	Internal - Confidential

Abstract. This document describes version 2 of the Nexora Platform API, which powers the NexaDesk, NexaFlow and NexaInsight products. It covers the service architecture, OAuth 2.0 authentication, the main resource endpoints, request limits by subscription plan, pagination, idempotency, webhooks and the error model. Latency objectives for each endpoint group are given so that client teams can set timeouts correctly.

Keywords: REST API, OAuth 2.0, JWT, rate limiting, pagination, webhooks, idempotency, error codes

1. Architecture Overview

All external traffic enters through the API Gateway, which terminates TLS 1.2+, applies rate limits stored in Redis, and routes requests to backend services running on Kubernetes (Amazon EKS). Access tokens are verified by the Auth Service. Persistent data is stored in PostgreSQL, domain events are published to Kafka and analytical data is stored in ClickHouse (Figure 1).

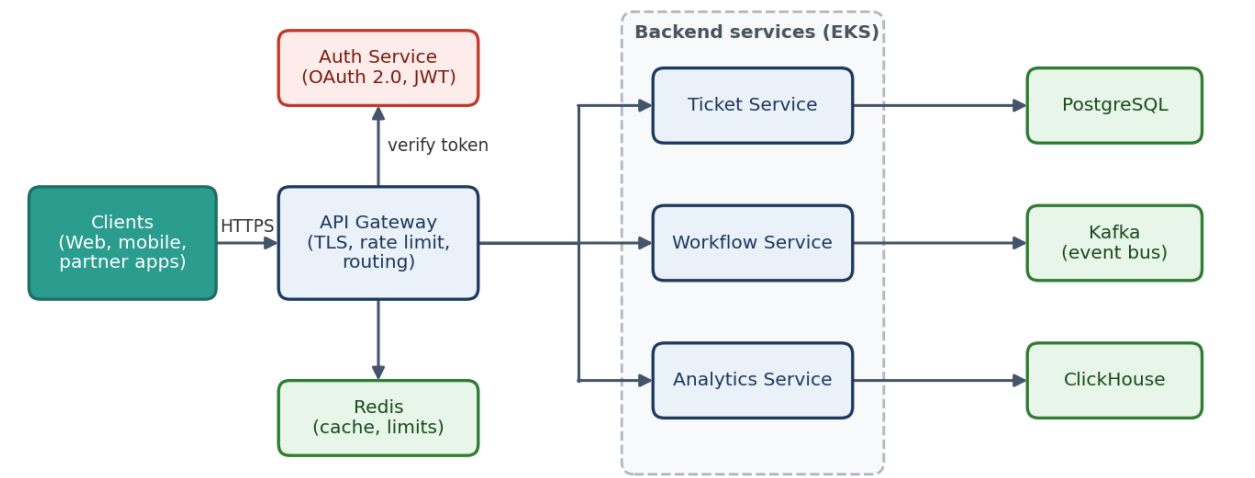


Figure 1. High-level architecture of the Nexora Platform API.

2. Base URL, Versioning and Conventions

- Base URL (production): `https://api.nexora.example/v2;` sandbox: `https://sandbox.api.nexora.example/v2.`
- All requests and responses use JSON (UTF-8). Timestamps are ISO 8601 in UTC, for example `2026-03-01T09:30:00Z`.
- Version 2 is the current version and is supported until **31 December 2027**. Version 1 was retired on 30 June 2026 and returns HTTP 410 (Gone).
- Field names use snake_case. Unknown fields in requests are rejected with HTTP 422.

3. Authentication

The API uses the OAuth 2.0 client-credentials grant. A client exchanges its `client_id` and `client_secret` for a JWT access token valid for **60 minutes**. The token is then sent in the `Authorization: Bearer` header. Secrets must be stored in a secrets manager and rotated at least every 90 days; they must never be committed to source control (see Coding Standards, NX-ENG-003).

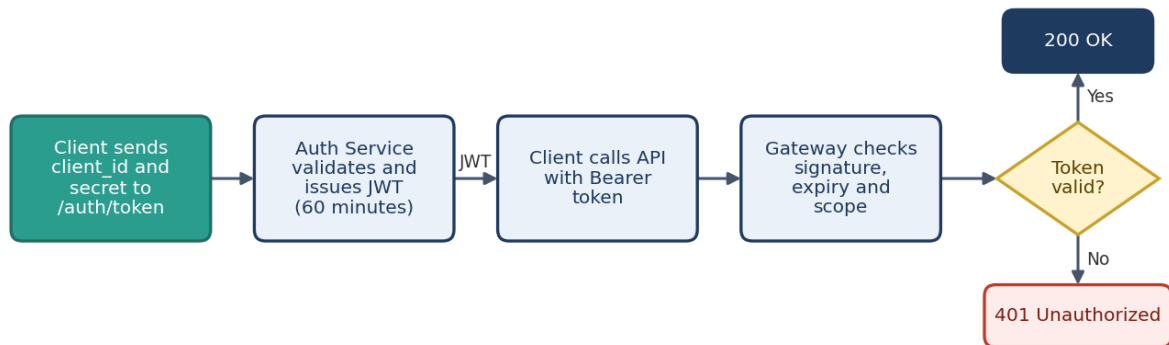


Figure 2. OAuth 2.0 client-credentials flow and token validation.

```
POST /v2/auth/token
Content-Type: application/json
```

```
{ "grant_type": "client_credentials",
  "client_id": "cli_8f2a91",
  "client_secret": "<stored-in-secrets-manager>",
  "scope": "tickets:read tickets:write" }
```

```
200 OK
{ "access_token": "eyJhbGciOiJIUzI1NiIs... ", "token_type": "Bearer", "expires_in": 3600 }
```

4. Endpoints

Table 1. Main resource endpoints of API v2.

Method	Path	Description	Scope
POST	/auth/token	Obtain an access token	-
GET	/tickets	List tickets (cursor pagination, filters: status, assignee, created_after)	tickets:read
POST	/tickets	Create a ticket; supports the Idempotency-Key header	tickets:write
GET	/tickets/{id}	Retrieve one ticket with its messages	tickets:read
PATCH	/tickets/{id}	Update status, priority, assignee or tags	tickets:write
DELETE	/tickets/{id}	Soft-delete a ticket (recoverable for 30 days)	tickets:write
GET	/customers	List customer records	customers:read
POST	/workflows/{id}/run	Trigger a NexaFlow workflow run	workflows:run
GET	/analytics/summary	Aggregated metrics (volume, resolution time, CSAT)	analytics:read
POST	/webhooks	Register a webhook endpoint	webhooks:manage

```
POST /v2/tickets
Authorization: Bearer <token>
Idempotency-Key: 6c1f2a3e-9d1b-4a77-9a55-1b0e7c2d9f10
```

```
{ "subject": "Cannot reset password",
  "customer_id": "cus_20411",
  "priority": "high",
  "channel": "email" }
```

```
201 Created
{ "id": "tkr_774120", "status": "open", "created_at": "2026-03-01T09:30:00Z" }
```

5. Rate Limits, Pagination and Idempotency

Limits apply per API client and depend on the customer's subscription plan (see Product Documentation, NX-PRD-001). Bursts of up to 2 times the limit are allowed for 10 seconds. Exceeding the limit returns HTTP 429 with a `Retry-After` header; clients should back off exponentially.

Table 2. Rate limits by subscription plan.

Plan	Requests per minute	Burst (10 s)	Webhook endpoints
Starter	60	120	3
Growth	300	600	10
Enterprise	1,200	2,400	50

- **Pagination:** cursor-based. Use `limit` (default 25, maximum 100) and pass `next_cursor` from the previous response.
- **Idempotency:** POST requests may include an Idempotency-Key; repeated requests with the same key within 24 hours return the original response.
- **Concurrency:** use the `If-Match` header with the ticket's ETag on PATCH to avoid lost updates (HTTP 409 on conflict).

6. Webhooks

Nexora sends an HTTPS POST to registered endpoints when events such as `ticket.created`, `ticket.updated` and `workflow.completed` occur. Every delivery carries an `X-Nexora-Signature` header containing an HMAC-SHA256 of the raw body using the webhook secret; receivers must verify the signature and respond with 2xx within 10 seconds. Failed deliveries are retried 5 times, after 1 minute, 5 minutes, 30 minutes, 2 hours and 12 hours.

7. Error Model

Table 3. HTTP status codes and Nexora error codes.

HTTP	Code	Meaning	Client action
400	NX-4000	Malformed request or invalid JSON	Fix the request
401	NX-4010	Missing, expired or invalid token	Request a new token
403	NX-4030	Token lacks the required scope	Request the correct scope
404	NX-4040	Resource does not exist	Check the identifier
409	NX-4090	Conflict (ETag mismatch or duplicate key)	Re-read and retry
422	NX-4220	Validation failed or unknown field	Correct field values
429	NX-4290	Rate limit exceeded	Wait for <code>Retry-After</code> , then retry
500	NX-5000	Unexpected server error	Retry with backoff; contact support with the <code>request_id</code>
503	NX-5030	Service temporarily unavailable	Retry with backoff

8. Performance Objectives

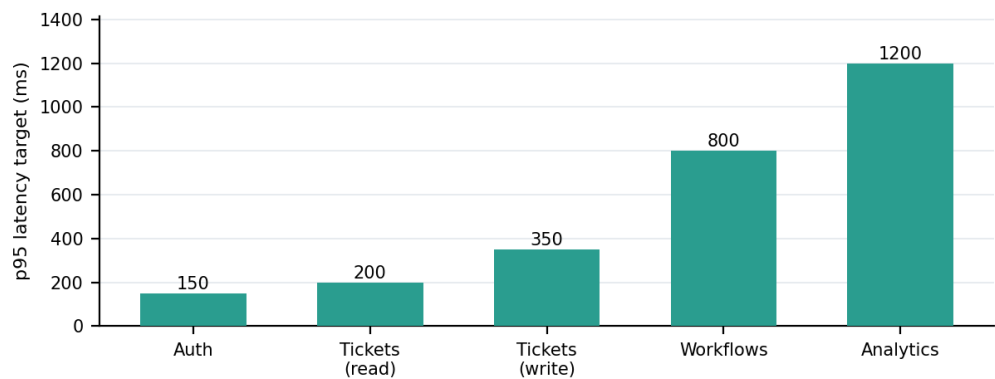


Figure 3. Target p95 response latency by endpoint group (milliseconds).

Clients should set a request timeout of at least 3 times the p95 target of the endpoint group they call. Platform availability targets are defined per subscription plan in the Product Documentation.

9. References

- [1] RFC 6749: The OAuth 2.0 Authorization Framework.
- [2] RFC 7519: JSON Web Token (JWT).
- [3] Nexora Deployment SOP, NX-ENG-002.
- [4] Nexora Coding Standards, NX-ENG-003 (secrets handling).
- [5] Nexora Product Documentation, NX-PRD-001 (plans and limits).

10. Revision History

Table 4. Document revision history.

Version	Date	Author	Summary of change
2.0	01 Sep 2025	Nexora Platform Engineering	Introduced v2 with cursor pagination and OAuth 2.0
2.3	15 Jan 2026	Nexora Platform Engineering	Added idempotency keys and ETag concurrency control
2.4	01 Mar 2026	Nexora Platform Engineering	Documented v1 retirement and updated rate limits